

Question	Answer	Marks
8(a)	<u>any three points from:</u> • <i>either:</i> photons contain discrete amounts of energy <i>or:</i> energy of photons depends on wavelength (of radiation) • photon energy equals difference between energy levels • discrete wavelengths (of spectral lines) means electron transitions must involve discrete energy changes • discrete energy changes means energy levels must be discrete	B3
8(b)(i)	galaxy moving away (from Earth) causes redshift	B1
	<u>observed</u> wavelength is greater (than emitted wavelength)	B1
8(b)(ii)	four lines drawn, each to the right of the corresponding line in Fig. 8.1	B1
	approximately equal separation of each drawn line from its corresponding original line	B1
8(b)(iii)	$\nu = H_0 d$ $= 2.1 \times 10^{-18} \times 6.7 \times 10^{24}$ $= 1.4(1) \times 10^7 \text{ m s}^{-1}$	C1
	$\Delta\lambda = (1.41 \times 10^7 \times 656) / (3.0 \times 10^8)$ $= 31 \text{ nm}$	C1
	wavelength = 656 + 31 = 687 nm	A1